

Predictive Stability Under Distribution Shift: Calibration, Discrimination, and Decision Thresholds with a Loan-Default Application

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Table of Contents

Abstract	2
1 Introduction	3
2 Literature Review and Statistical Background	5
2.1 Traditional and machine-learning credit scoring	5
2.2 Class imbalance and threshold selection	5
2.3 Probability quality, calibration, and discrimination	5
2.4 Distribution shift and predictive stability	6
2.5 Fairness, interpretability, and responsible use	6
3 Statistical and Computational Framework	7
4 Data	8
4.1 Predictor Construction	8
5 Methods	9
5.1 Development Design	9
5.2 Leakage-Safe Preprocessing	9
5.3 Candidate Models	9
5.4 Model Fitting and Optimization	10
5.5 Deployment-Stability Simulation Design	10
6 Results	13
6.1 Validation-Based Model Selection	13
6.2 Decision-Threshold Optimization	14
6.3 Held-Out Test Performance	16
6.4 Discrimination: ROC and Precision–Recall Performance	17
6.5 Calibration	18
6.6 Risk Stratification	20
6.7 Predictor Reliance	20
6.8 Age-Related Subgroup and Sensitivity Diagnostics	21
6.9 Separate Validation / Stress Test	23
6.10 Predictive Stability Under Controlled Distribution Shift	23

7 Discussion	29
8 Limitations	31
9 Future Research	32
10 Reproducibility and Software	33
11 Conclusion	34
References	35

Abstract

A probability model can perform well during internal validation yet behave differently when the population in which it is deployed differs from the population in which it was developed. This study investigates that problem through a unified empirical and simulation analysis of loan-default prediction. The empirical component uses 255,347 loan records with 29,653 recorded defaults (approximately 11.6%) to establish a probability model, assess calibration, and select an operating threshold. Logistic regression, ridge logistic regression, random forest, and gradient boosting are compared using validation log loss as the primary probability-quality criterion. Raw logistic regression achieved the lowest validation log loss and, on the locked test set, achieved ROC-AUC 0.7585, PR-AUC 0.3288, Brier score 0.09085, and log loss 0.31111. A validation-selected balanced-accuracy threshold of 0.1175 produced held-out recall of 68.4% and specificity of 70.1%.

The central research question is whether those statistical properties remain stable when the deployment distribution changes. Because the empirical data do not contain documented temporal cohorts or an independent deployment population, controlled Monte Carlo experiments are used to isolate covariate shift, intercept changes in the conditional outcome mechanism, and changes in predictor effects. Across 500 replications per condition, pure covariate shift produced little excess probability error when the conditional logistic model was correctly specified, whereas covariate shift combined with omitted nonlinear structure produced substantial calibration and probability deterioration. Under intercept outcome shift, calibration changed markedly while model ranking remained close to the oracle. Under slope outcome shift, the original model’s ROC-AUC declined from approximately 0.762 to 0.473 as the predictor–outcome relationship reversed, while the oracle remained informative. Decision-threshold experiments also showed that the balanced-accuracy-optimal threshold can change substantially after population shift even when the conditional probability model remains correctly specified.

The results show that discrimination, calibration, probability accuracy, and decision-threshold stability are distinct properties of a predictive system. Internal predictive performance should therefore not be interpreted as evidence of deployment stability without temporal or external validation. Because the recorded `Default` label has no documented observation horizon, the empirical estimates are interpreted as risk of the recorded outcome rather than formal 12-month or lifetime probabilities of default.

Keywords: loan default; distribution shift; covariate shift; logistic regression; probability calibration; model misspecification; decision threshold; Monte Carlo simulation

1 Introduction

Loan-default prediction is fundamentally a problem of statistical prediction under uncertainty. A fitted model estimates the probability that a borrower experiences a recorded default outcome,

$$p_i = P(Y_i = 1 \mid X_i),$$

and a later decision rule may convert the estimated probability $\hat{p}_i$ into an operational action. The statistical validity of that system therefore depends on more than whether the model can classify observations correctly. The estimated probabilities should be appropriately calibrated, the model should rank higher-risk observations above lower-risk observations, and the operating threshold should remain appropriate for the population in which the model is used.

These properties are usually evaluated using held-out observations drawn from the same underlying source as the development data. That design is useful for internal validation but leaves an important question unresolved: **what happens when the deployment population no longer follows the same distribution as the development population?** Changes in borrower characteristics, underwriting practices, macroeconomic conditions, product mix, or the relationship between predictors and default can alter the statistical environment in which a model operates.

This distinction can be written formally. Let $P(X, Y)$ denote the development joint distribution and $Q(X, Y)$ the deployment joint distribution. Distribution shift occurs when

$$P(X, Y) \neq Q(X, Y).$$

The consequences depend on which part of the joint distribution changes. Under covariate shift,

$$P(X) \neq Q(X), \quad P(Y \mid X) = Q(Y \mid X),$$

whereas a change in the conditional outcome mechanism occurs when

$$P(Y \mid X) \neq Q(Y \mid X).$$

These mechanisms need not affect a predictive model in the same way. A model may preserve ranking while losing calibration, remain probabilistically accurate while its operating threshold changes, or lose discrimination when predictor–outcome relationships themselves change.

The loan-default application in this paper provides an empirical baseline for studying these questions. The dataset contains a minority default outcome, making overall accuracy alone unsuitable for model assessment. Candidate probability models are compared using validation log loss, the selected model is assessed for discrimination and calibration, and a review threshold is selected independently using balanced accuracy. The empirical analysis therefore establishes the predictive system whose components are of interest: probability estimation, calibration, ranking, and threshold-based decision performance.

The empirical data, however, contain no documented temporal cohorts or genuinely independent deployment population. They cannot directly reveal how those components would respond to future population change. Controlled Monte Carlo simulation is therefore used as the second methodological component of the same study. The simulations hold some parts of the data-generating process fixed while deliberately changing others, allowing the effects of different forms of distribution shift to be isolated.

The study addresses five related questions:

1. What level of internal probability performance, discrimination, and calibration is achieved by the empirical loan-default model selected under a prespecified validation criterion?
2. Under covariate shift, when does a fitted probability model remain stable, and when does model misspecification become increasingly consequential?
3. Can discrimination remain stable or improve while probability calibration deteriorates?
4. How does changing a predictor–outcome relationship affect discrimination and calibration when event prevalence is held approximately constant?
5. Can a decision threshold selected in the development population become inappropriate after population shift even when the underlying probability model remains useful?

The contribution is not a claim that one algorithm is universally superior for credit risk, nor is the simulation intended to establish that a particular form of shift has occurred in the observed loan data. Instead, the paper uses a real prediction problem and controlled statistical experiments to study a broader question: **which properties of a probability-based predictive system remain stable under distributional change, and which must be reassessed?**

2 Literature Review and Statistical Background

2.1 Traditional and machine-learning credit scoring

Logistic regression remains a natural benchmark for binary credit outcomes because it models a conditional probability through the log-odds scale and produces interpretable coefficients. At the same time, contemporary credit-scoring research increasingly evaluates random forests, boosting, neural networks, and hybrid approaches that can represent nonlinear effects and higher-order interactions. Montevechi et al. (2024) review machine learning in consumer credit-risk modelling and emphasize both its potential and the need for standardized modelling procedures and interpretability. Ayari, Guetari, and Kraïem (2026), reviewing recent financial credit-scoring research, similarly identify predictive performance, dimensionality, interpretability, bias, and deployment complexity as recurring issues.

The existence of more flexible algorithms does not imply that they must outperform logistic regression in every dataset. Bazzana, Bee, and Hussin Adam Khatir (2024), for example, report that machine-learning methods produced only modest gains over logistic regression in their default-prediction application and argue that these gains must be weighed against implementation and interpretation costs. That observation motivates the validation-based comparison used here: complexity is treated as a candidate capability, not as evidence of superiority.

2.2 Class imbalance and threshold selection

In an imbalanced credit portfolio, a 0.50 cutoff is not automatically appropriate. Bazzana, Bee, and Hussin Adam Khatir (2024) explicitly discuss threshold adjustment for default prediction because sensitivity and specificity cannot generally be maximized simultaneously. The statistical-learning principles make the same conceptual point more generally: fitting the conditional probability and choosing the final classification threshold are separate operations, and a different threshold can be justified when the consequences of the two types of classification error differ.

This paper uses balanced accuracy as the default threshold objective because verified monetary costs of false negatives and false positives are not available. Balanced accuracy gives equal weight to sensitivity and specificity. It is therefore a statistical operating objective, not a claim of economic optimality. The analysis also reports alternative validation-selected thresholds for F1, F2, and Youden’s J so that the consequences of changing the objective are visible.

2.3 Probability quality, calibration, and discrimination

Binary risk models should be evaluated along more than one dimension. Steyerberg et al. (2010) emphasize that discrimination, calibration, and overall predictive performance answer different questions. ROC-AUC measures ranking ability across thresholds, while precision-recall performance is especially informative when the positive class is less common. The Brier score summarizes squared probability error, and log loss penalizes probability assignments through the negative Bernoulli log-likelihood.

Calibration asks whether predicted probabilities correspond to observed frequencies. Niculescu-Mizil and Caruana (2005) showed that different learning algorithms can produce systematically distorted probability scales and studied post-hoc methods including Platt scaling. Accordingly, this study does not assume that calibration must improve a model; raw and Platt-calibrated probabilities are compared on held-out validation data.

2.4 Distribution shift and predictive stability

Internal validation estimates performance under the observed development environment. Deployment can violate that environment when the joint distribution of predictors and outcomes changes. Under covariate shift, the marginal predictor distribution changes while the conditional outcome mechanism remains invariant. In principle, a correctly specified conditional model can remain valid under such a shift because $P(Y | X)$ is unchanged. Under misspecification, however, the approximation learned in the development population can become less accurate when deployment places more probability mass in regions where the approximation is poor (Shimodaira, 2000; Quiñonero-Candela et al., 2009; Wen, Yu, and Greiner, 2014).

A change in $P(Y | X)$ represents a different problem. An intercept change in a logistic outcome mechanism can alter absolute risk while preserving much of the ranking induced by the original linear predictor. A change in predictor slopes can alter the ranking itself. These distinctions motivate separate assessment of discrimination, calibration, probability loss, and threshold performance rather than relying on a single metric such as ROC-AUC.

2.5 Fairness, interpretability, and responsible use

Credit scoring is a high-consequence application in which predictive performance is not the only consideration. Kozodoi, Jacob, and Lessmann (2022) show that fairness criteria can interact with profitability and model design, while Das, Stanton, and Wallace (2023) review algorithmic fairness with particular attention to credit scoring. These studies motivate the subgroup diagnostics reported later in this paper. The available dataset does not contain the full set of attributes needed for a complete fairness audit, so the subgroup analysis is descriptive and should not be interpreted as establishing that the model is fair.

3 Statistical and Computational Framework

The analysis treats a predictive system as a sequence of connected but distinct statistical components:

Probability model $\rightarrow$ Calibration $\rightarrow$ Ranking $\rightarrow$ Decision threshold $\rightarrow$ Deployment stability

The empirical analysis estimates and evaluates the first four components under internal validation. The simulation analysis then studies the fifth component by changing the deployment data-generating mechanism while monitoring the same probability, calibration, ranking, and threshold quantities.

The statistical organization follows the factorization

$$P(X, Y) = P(Y | X)P(X).$$

This decomposition separates changes in the population of predictors from changes in the conditional outcome mechanism. The simulation experiments are therefore not separate applications: they are controlled perturbations of the same joint distribution underlying the empirical prediction problem. Covariate shift changes $P(X)$ while holding $P(Y | X)$ fixed; the outcome-shift experiments change $P(Y | X)$. For each perturbation, the analysis asks how four observable properties respond:

probability accuracy, calibration, discrimination, decision-rule stability.

The analysis is organized around the progression

statistical question $\rightarrow$ representation $\rightarrow$ model $\rightarrow$ objective/loss $\rightarrow$ computation $\rightarrow$ validation $\rightarrow$ decision.

Five principles guide the implementation:

1. **Representation matters.** Raw predictors must be converted into a numerical representation appropriate to the model.
2. **Loss defines the optimization target.** Logistic regression maximizes Bernoulli likelihood, while validation log loss evaluates probability quality on observations not used to estimate the model.
3. **Computation should be stable and reproducible.** Scaling and training-only preprocessing are used to avoid unnecessary numerical and leakage problems.
4. **Flexible models require validation.** Additional complexity is retained only when it improves held-out evidence.
5. **Probability fitting and classification are separate.** The threshold is optimized only after the probability model has been selected.

Decision objectives are evaluated over a grid and visualized so that the resulting trade-offs remain transparent.

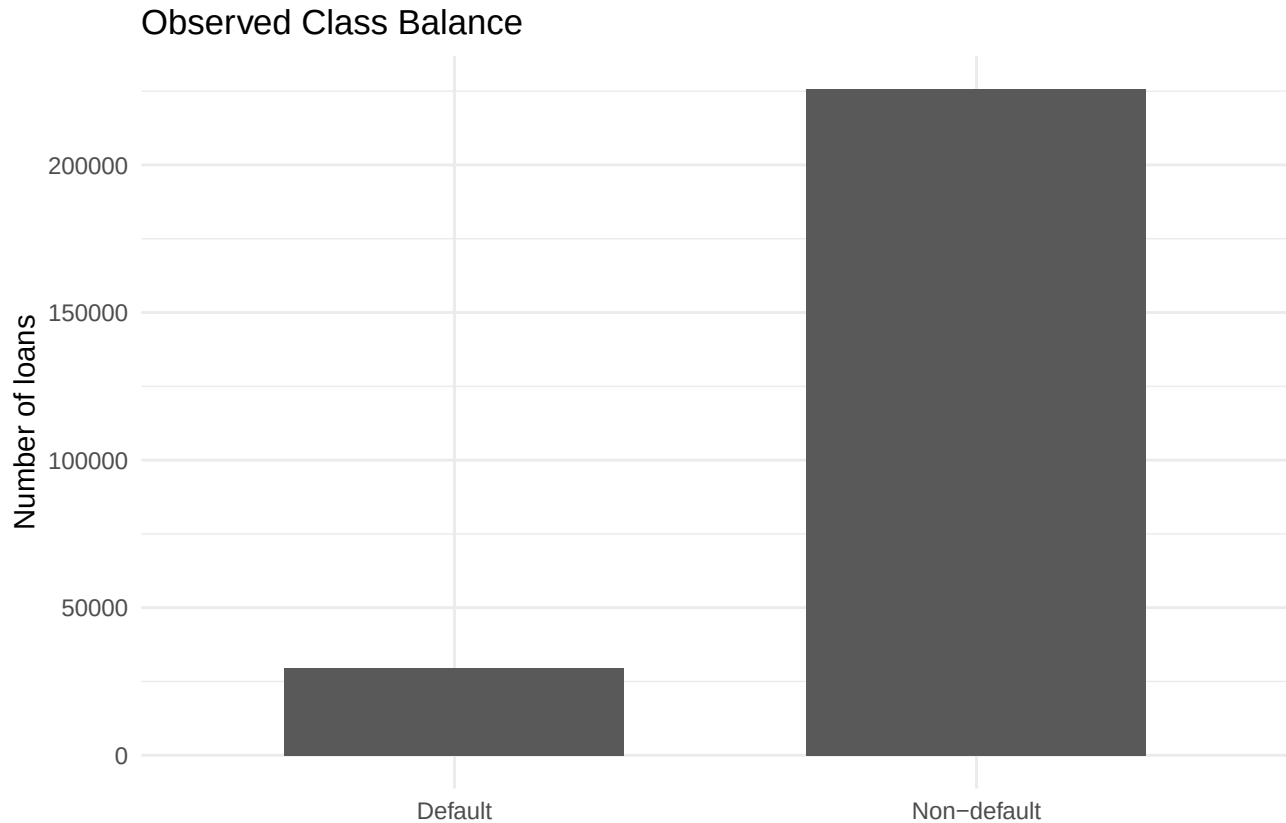
4 Data

Table 1: Dataset overview.

Loans	Predictors	Defaults	DefaultRate	MissingCells
255347	16	29653	11.61%	0

`LoanID` is an identifier and is not used as a predictor.

The study data are tabular loan-level observations. There is no observed repayment sequence, default date, recovery history, or time-varying balance. Therefore, recurrent or sequence models are not appropriate representations of the information available here.



Because default is the minority outcome, raw accuracy can be misleading. Probability loss, ranking metrics, calibration, recall, precision, specificity, and review rate are examined instead.

4.1 Predictor Construction

The engineered ratio

$$\text{LoanToAnnualIncome} = \frac{\text{LoanAmount}}{\text{Income}}$$

captures loan size relative to annual income. It is derived only from variables already present in the data.

5 Methods

5.1 Development Design

The analysis uses four non-overlapping partitions:

- **60% training:** fit preprocessing and model parameters;
- **15% calibration:** fit optional probability calibration;
- **10% validation:** select the probability model and the threshold rule;
- **15% test:** final evaluation after choices are locked.

Table 2: Development partitions.

Subset	Loans	DefaultRate
train	153207	11.61%
calibration	38301	11.61%
validation	25534	11.61%
test	38305	11.62%

This design assumes each row is an independent loan. If repeated borrower IDs or origination dates become available, a borrower-grouped or out-of-time design would be preferable.

5.2 Leakage-Safe Preprocessing

Preprocessing is learned **only on the training subset**.

Standardization is useful for regularized optimization because predictors on very different numerical scales can create poor conditioning and unstable numerical behavior.

5.3 Candidate Models

The candidate set contains models with different levels of flexibility.

Model	Statistical role
Logistic regression	Interpretable linear log-odds baseline
Ridge logistic regression	Logistic model with coefficient shrinkage
Random forest	Nonlinear tree ensemble
Gradient boosting	Sequential nonlinear ensemble
Neural network (optional)	Flexible nonlinear composition

The purpose of including more complex models is not to assume they are superior. Flexible models are retained only when they demonstrate stronger held-out predictive performance, because additional flexibility can also increase overfitting risk.

5.3.1 Probability-Loss Criterion

For a binary outcome, the validation log loss is

$$-\frac{1}{n} \sum_{i=1}^n [y_i \log(\hat{p}_i) + (1 - y_i) \log(1 - \hat{p}_i)].$$

This is the negative average Bernoulli log-likelihood. Because this project is intended to estimate probabilities, log loss is the **primary model-selection criterion**.

Brier score, ROC-AUC, and PR-AUC are used as supporting diagnostics.

5.4 Model Fitting and Optimization

Statistical estimation can be expressed as an optimization problem. Logistic regression optimizes Bernoulli likelihood; ridge regression adds a penalty; boosting and neural networks optimize loss numerically.

5.5 Deployment-Stability Simulation Design

The empirical analysis provides an internally validated reference model but cannot identify how model behavior changes under a genuinely different deployment population. A controlled Monte Carlo design is therefore used to study distribution shift while keeping the relevant data-generating mechanisms known.

5.5.1 Baseline data-generating mechanism

Three independent standardized predictors are generated as

$$X_1, X_2, X_3 \stackrel{\text{iid}}{\sim} N(0, 1),$$

with binary outcome

$$Y_i \mid X_i \sim \text{Bernoulli}(p_i),$$

and baseline logistic mechanism

$$\text{logit}(p_i) = \beta_0 + 0.8X_{i1} - 0.6X_{i2} + 0.4X_{i3}.$$

The intercept is selected so that baseline event prevalence is approximately 12%, close to the empirical default rate. The slopes 0.8, -0.6 , and 0.4 are simulation design parameters chosen to create moderate effects of different magnitudes and directions; they are not estimates from the empirical loan data.

Each condition uses $R = 500$ Monte Carlo replications with simulation seed 1998. Unless otherwise stated, training, validation/adaptation, and test samples contain 5,000 observations. Fresh samples are generated within each replication. The fitted model is ordinary logistic regression.

Before introducing shift, a coefficient-recovery experiment verifies the implementation under correct specification. Across 500 replications, the mean estimates were -2.3903 , 0.8030 , -0.6028 , and 0.3991 for true values -2.387 , 0.8 , -0.6 , and 0.4 , respectively.

5.5.2 Probability, discrimination, and calibration measures

Because the true data-generating probabilities are known in simulation, the fitted model can be compared with an oracle using the true deployment probabilities. Probability accuracy is summarized by log loss, Brier score, and

$$\text{PMSE} = \frac{1}{n} \sum_{i=1}^n (\hat{p}_i - p_i^{\text{true}})^2.$$

Excess log loss and excess Brier score are calculated relative to the oracle so that changes in population composition or intrinsic prediction difficulty are not confused with model error.

Discrimination is measured by ROC-AUC. Calibration is summarized by the logistic calibration model

$$\text{logit}\{P(Y = 1)\} = a + b \text{logit}(\hat{p}),$$

where ideal calibration corresponds to intercept $a = 0$ and slope $b = 1$.

5.5.3 Covariate shift and misspecification

The first experiment changes the deployment distribution of X_1 :

$$X_1^{\text{deploy}} \sim N(\delta, 1), \quad \delta \in \{0, 0.5, 1, 1.5, 2\},$$

while X_2 and X_3 remain standard normal. The conditional outcome mechanism is held fixed, so this is pure covariate shift.

To examine the interaction between shift and model misspecification, the true mechanism is extended to

$$\text{logit}(p) = \beta_0 + 0.8X_1 - 0.6X_2 + 0.4X_3 + \beta_4X_1^2,$$

with

$$\beta_4 \in \{0, 0.25, 0.50\}.$$

For each value of β_4 , the intercept is recalibrated so that baseline prevalence at $\delta = 0$ remains approximately 12%. The fitted logistic model deliberately omits X_1^2 . Thus $\beta_4 = 0$ is the correctly specified control condition, while 0.25 and 0.50 represent moderate and stronger misspecification.

5.5.4 Decision-threshold stability

For each misspecification condition and replication, the fitted model is trained on the baseline population. An independent baseline validation sample at $\delta = 0$ selects the threshold maximizing balanced accuracy over the prespecified grid 0.01, 0.015, ..., 0.50. For each deployment value of δ , a separate shifted adaptation sample estimates the deployment-optimal threshold, and an independent shifted test sample evaluates both the frozen baseline threshold and the adapted threshold.

This design separates threshold selection from threshold evaluation. The main quantities are threshold movement and the balanced-accuracy gain from adaptation.

5.5.5 Intercept outcome shift

The second outcome-mechanism experiment leaves the predictor distribution and slopes unchanged but adds a deployment intercept shift,

$$\text{logit}(p_{\text{deploy}}) = \beta_0 + \gamma + 0.8X_1 - 0.6X_2 + 0.4X_3,$$

where

$$\gamma \in \{0, 0.25, 0.50, 0.75, 1\}.$$

The original fitted model is not updated. Because

$$\text{logit}(p_{\text{deploy}}) = \gamma + \text{logit}(p_{\text{baseline}}),$$

the population calibration relationship is expected to have intercept $a = \gamma$ and slope $b = 1$.

5.5.6 Slope outcome shift

The final experiment changes the true deployment effect of X_1 :

$$\beta_{1,\text{deploy}} = 0.8 - \kappa,$$

with

$$\kappa \in \{0, 0.4, 0.8, 1.2, 1.6\}.$$

The deployment coefficient therefore moves from 0.8 to 0.4, 0, -0.4 , and -0.8 . For each condition, the deployment intercept is recalibrated to keep event prevalence near 12%. This isolates deterioration caused by a changing predictor–outcome relationship rather than by a large prevalence change.

5.5.7 Monte Carlo uncertainty

For a simulation summary $\bar{\theta}$, Monte Carlo standard error is

$$\text{MCSE}(\bar{\theta}) = \frac{s_{\theta}}{\sqrt{R}},$$

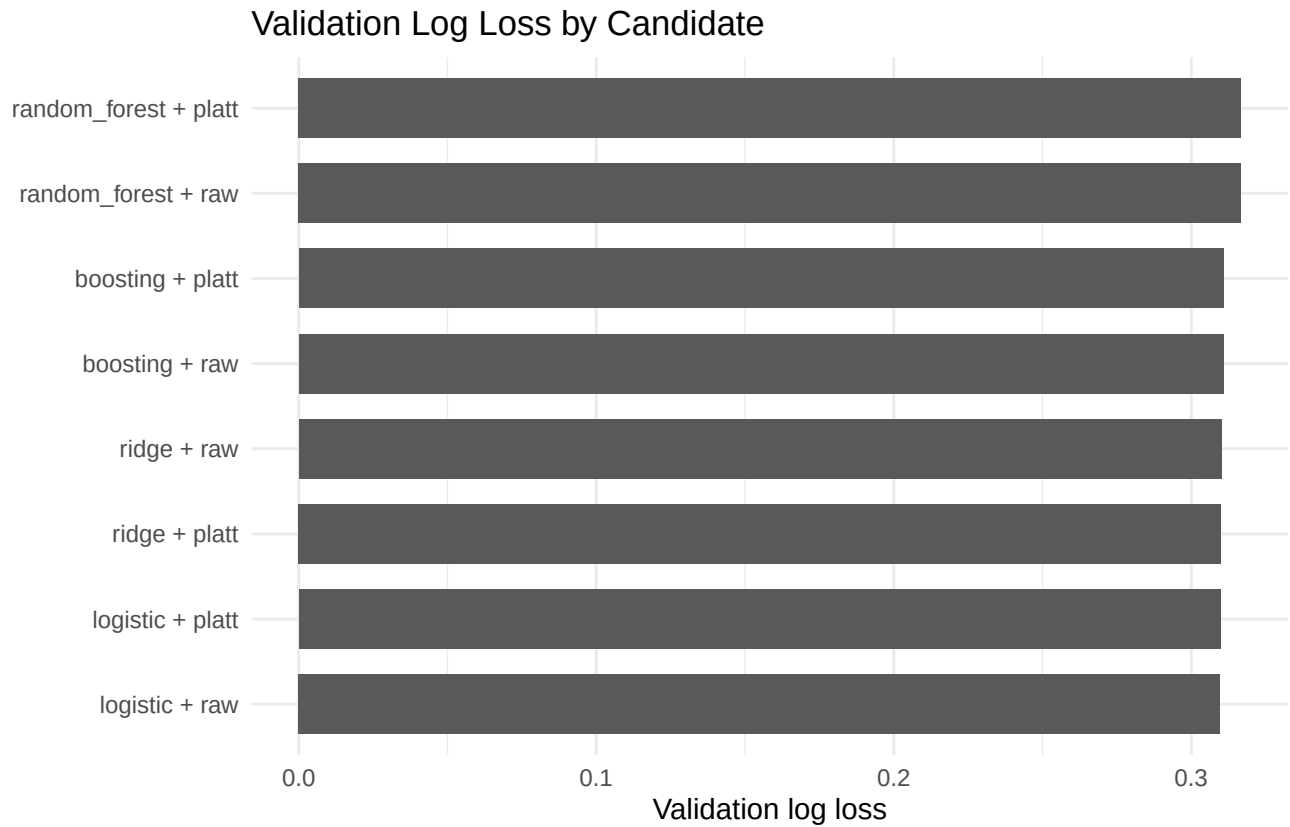
where s_{θ} is the empirical standard deviation across the $R = 500$ replications. MCSE quantifies numerical uncertainty from the finite number of simulation runs; it is not an uncertainty interval for real-world deployment performance.

6 Results

6.1 Validation-Based Model Selection

Table 3: Validation probability performance. Lower LogLoss and Brier are better; higher ROC-AUC and PR-AUC are better.

Model	Calibration	LogLoss	Brier	ROC_AUC	PR_AUC
Logistic regression	Raw	0.3097	0.0905	0.7620	0.3320
Logistic regression	Platt	0.3098	0.0905	0.7620	0.3320
Ridge logistic	Platt	0.3099	0.0906	0.7615	0.3318
Ridge logistic	Raw	0.3101	0.0906	0.7615	0.3318
Gradient boosting	Raw	0.3108	0.0908	0.7603	0.3281
Gradient boosting	Platt	0.3108	0.0908	0.7603	0.3281
Random forest	Raw	0.3167	0.0922	0.7443	0.3086
Random forest	Platt	0.3168	0.0922	0.7443	0.3086



The lowest validation log loss was obtained by **logistic** with **raw** probabilities.

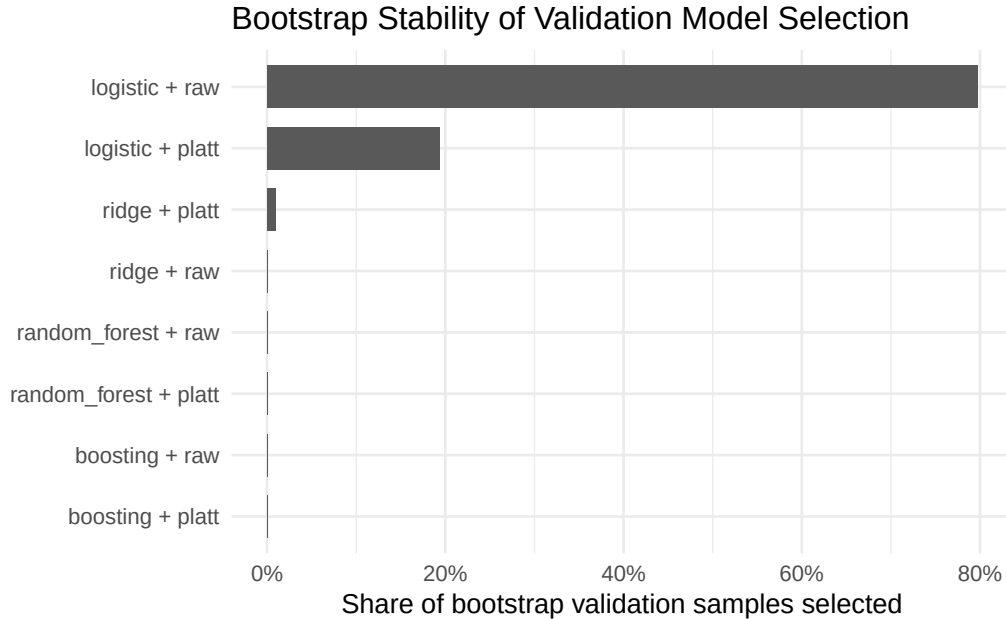
Selection is based primarily on validation log loss because the target of interest is a probability. A prespecified parsimony rule is used when candidates are practically tied within 0.001 log-loss units: the simpler candidate is preferred when additional complexity does not produce a meaningful validation improvement.

6.1.1 Bootstrap Stability of the Validation Ranking

A single validation table reports one realized comparison. To assess whether the ranking is fragile to the particular validation observations, the fitted candidate models are held fixed and the validation set is bootstrap-resampled. This is a **conditional stability analysis of the validation ranking**; it is not a substitute for a completely new external sample or repeated refitting on new development splits.

Table 4: Bootstrap stability of the validation model ranking, conditional on the fitted candidate models.

Candidate	Selection rate	Mean log loss	SD
logistic + raw	79.7%	0.30983	0.00379
logistic + platt	19.3%	0.30988	0.00376
ridge + platt	1.0%	0.31002	0.00377
ridge + raw	0.0%	0.31020	0.00374
boosting + raw	0.0%	0.31089	0.00379
boosting + platt	0.0%	0.31094	0.00376
random_forest + raw	0.0%	0.31683	0.00386
random_forest + platt	0.0%	0.31687	0.00384



A high bootstrap selection rate for the retained candidate indicates that its validation advantage is not driven only by a small number of observations. A diffuse distribution of selection rates would instead indicate that the candidate models are practically difficult to distinguish with the available validation evidence.

6.2 Decision-Threshold Optimization

Probability estimation and classification are treated as different steps. A different threshold can be appropriate when the consequences of false positives and false negatives differ.

For comparison, an F2-based objective is also evaluated. Because F2 weights recall more strongly than precision, it generally favors a lower cutoff and a larger review set.

The primary threshold objective is **balanced accuracy** because verified economic costs are unavailable:

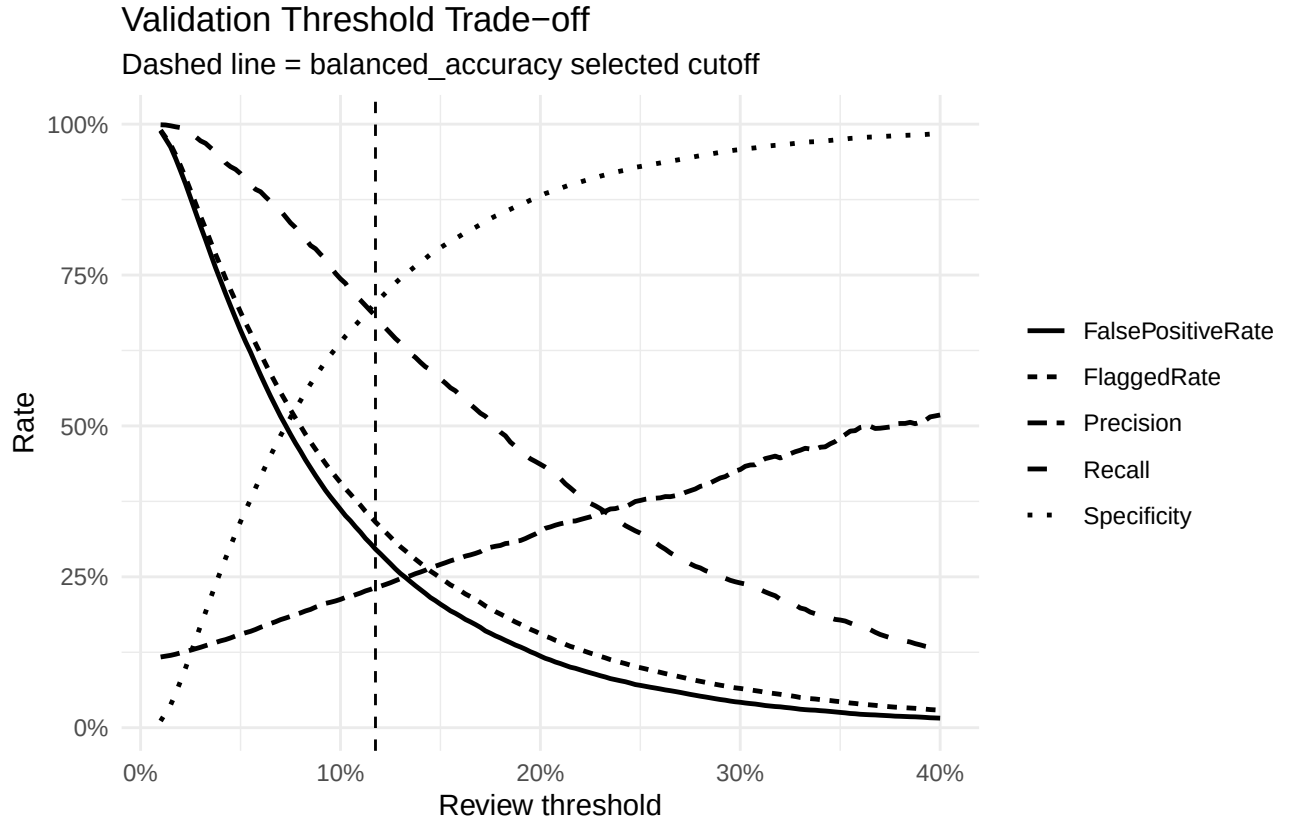
$$\text{Balanced Accuracy} = \frac{\text{Recall} + \text{Specificity}}{2},$$

which treats sensitivity and specificity symmetrically.

This should not be interpreted as a universal lending-policy optimum. If real false-negative and false-positive costs become available, expected cost should replace a generic statistical objective.

Table 5: Validation-selected thresholds under different statistical objectives.

Objective	Threshold
F2	9.2%
F1	17.2%
Balanced accuracy	11.8%
Youden J	11.8%



The chosen threshold is **11.8%** under the declared **balanced_accuracy** objective.

The threshold was selected **only on validation data**. The test set was not used to tune it.

6.2.1 Sensitivity to the Threshold Objective

Balanced accuracy is the primary threshold objective because verified monetary error costs are unavailable. To show that the operating rule depends on the objective rather than on the probability

model, four thresholds are selected independently on validation data and then evaluated on the locked test set.

Table 6: Test sensitivity, precision, and specificity under validation-selected threshold objectives.

Objective	Threshold	Recall	Precision	Specificity
F2	0.0925	0.7701	0.2021	0.6003
F1	0.1725	0.5038	0.2885	0.8367
Balanced accuracy	0.1175	0.6843	0.2313	0.7011
Youden J	0.1175	0.6843	0.2313	0.7011

Table 7: Test review burden and overall accuracy under validation-selected threshold objectives.

Objective	FalsePositiveRate	FlaggedRate	Accuracy
F2	0.3997	0.4428	0.6200
F1	0.1633	0.2028	0.7980
Balanced accuracy	0.2989	0.3437	0.6991
Youden J	0.2989	0.3437	0.6991

This comparison is not intended to identify a universally optimal cutoff. It demonstrates that the operating point depends on the chosen loss or utility criterion. In a production setting, verified economic costs or review-capacity constraints should replace a generic statistical threshold objective.

6.3 Held-Out Test Performance

Table 8: Held-out test probability performance.

Model	LogLoss	Brier	ROC_AUC	PR_AUC
logistic + raw	0.3111	0.0908	0.7585	0.3288

Table 9: Held-out test operating characteristics at the validation-selected threshold.

Threshold	Recall	Precision	Specificity	FalsePositiveRate	FlaggedRate	Accuracy
0.1175	0.6843	0.2313	0.7011	0.2989	0.3437	0.6991

Table 10: Held-out test confusion-matrix counts.

TP	FP	TN	FN
3045	10120	23735	1405

This table is the final generalization assessment for the selected probability model and validation-selected threshold.

6.3.1 Bootstrap Uncertainty for Final Test Metrics

Point estimates can conceal sampling uncertainty. The following intervals bootstrap the **locked test observations** while holding the fitted model and threshold fixed. They quantify uncertainty in

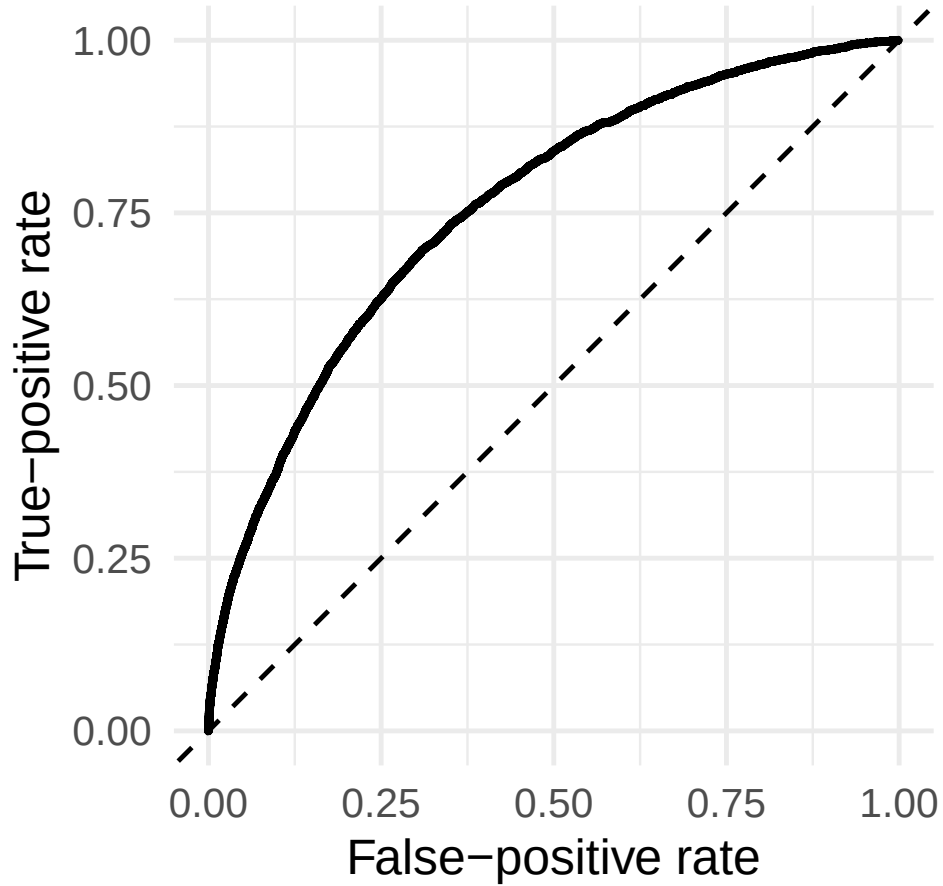
the reported test metrics; they do not account for uncertainty arising from an entirely new model-development sample.

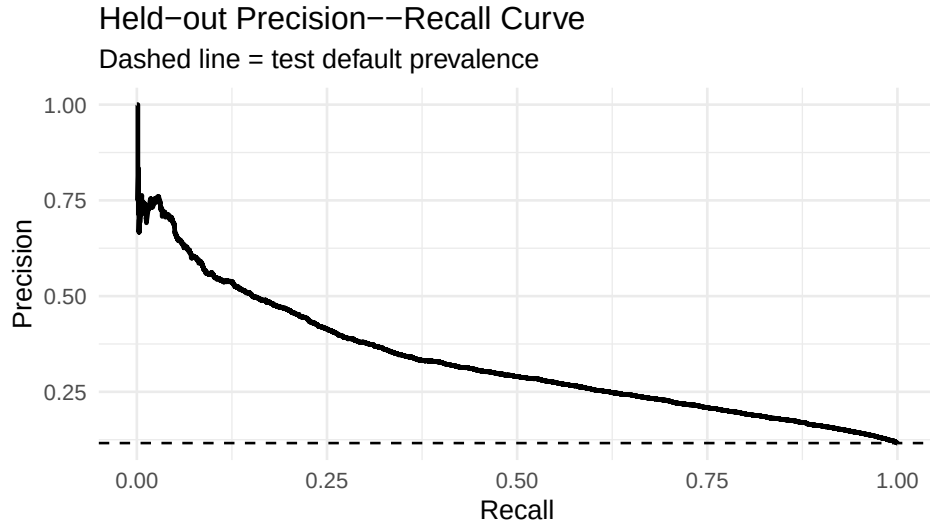
Table 11: Bootstrap 95% intervals for locked test performance.

Metric	Estimate	Lower 95%	Upper 95%
ROC_AUC	0.7585	0.7518	0.7651
PR_AUC	0.3288	0.3147	0.3423
Brier	0.0908	0.0887	0.0930
LogLoss	0.3111	0.3050	0.3170
Recall	0.6843	0.6718	0.6976
Precision	0.2313	0.2242	0.2387
Specificity	0.7011	0.6959	0.7056
FalsePositiveRate	0.2989	0.2944	0.3041
FlaggedRate	0.3437	0.3394	0.3485
Accuracy	0.6991	0.6945	0.7034

6.4 Discrimination: ROC and Precision–Recall Performance

Held-out ROC Curve



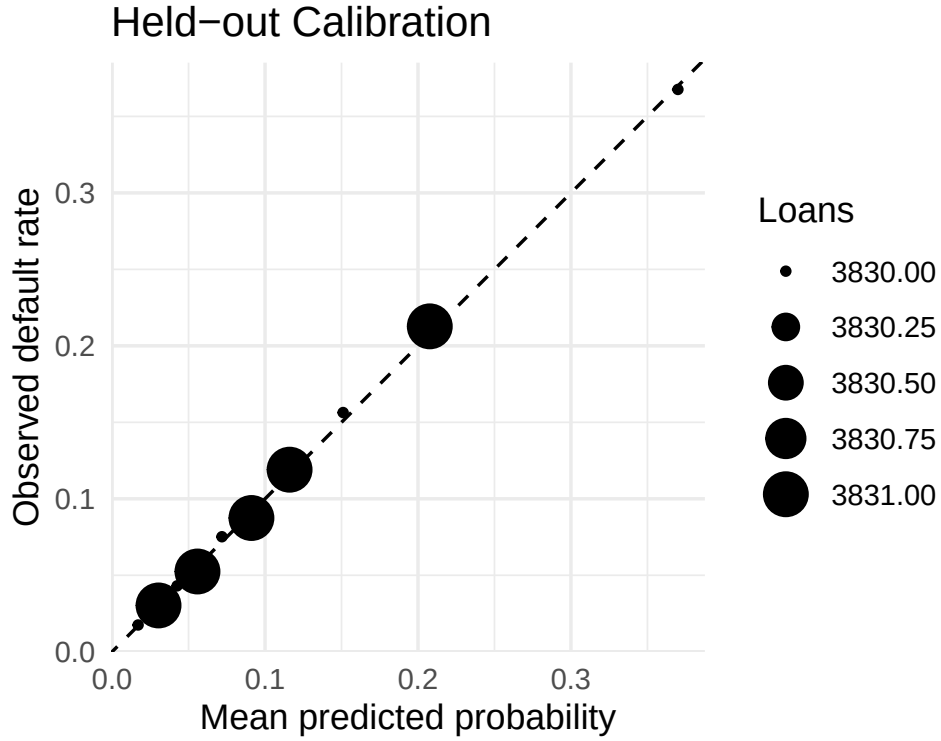


6.5 Calibration

A useful risk model should not only rank loans correctly; its probabilities should also correspond reasonably to observed frequencies.

Table 12: Held-out calibration by risk group.

Risk group	N	Mean predicted	Observed default rate
$[-\text{Inf}, 0.0241]$	3830	0.0170	0.0175
$(0.0241, 0.0364]$	3831	0.0304	0.0303
$(0.0364, 0.049]$	3830	0.0425	0.0431
$(0.049, 0.0633]$	3831	0.0558	0.0525
$(0.0633, 0.0808]$	3830	0.0718	0.0752
$(0.0808, 0.102]$	3831	0.0911	0.0874
$(0.102, 0.131]$	3831	0.1161	0.1190
$(0.131, 0.174]$	3830	0.1510	0.1564
$(0.174, 0.251]$	3831	0.2077	0.2127
$(0.251, \text{Inf}]$	3830	0.3699	0.3676



6.5.1 Calibration Intercept and Slope

The grouped calibration plot provides a visual diagnostic. Two additional summaries are estimated on the locked test set. A calibration intercept near 0 indicates correct average risk, while a calibration slope near 1 indicates that the spread of predicted risks is appropriate.

Table 13: Calibration intercept and slope on the locked test set.

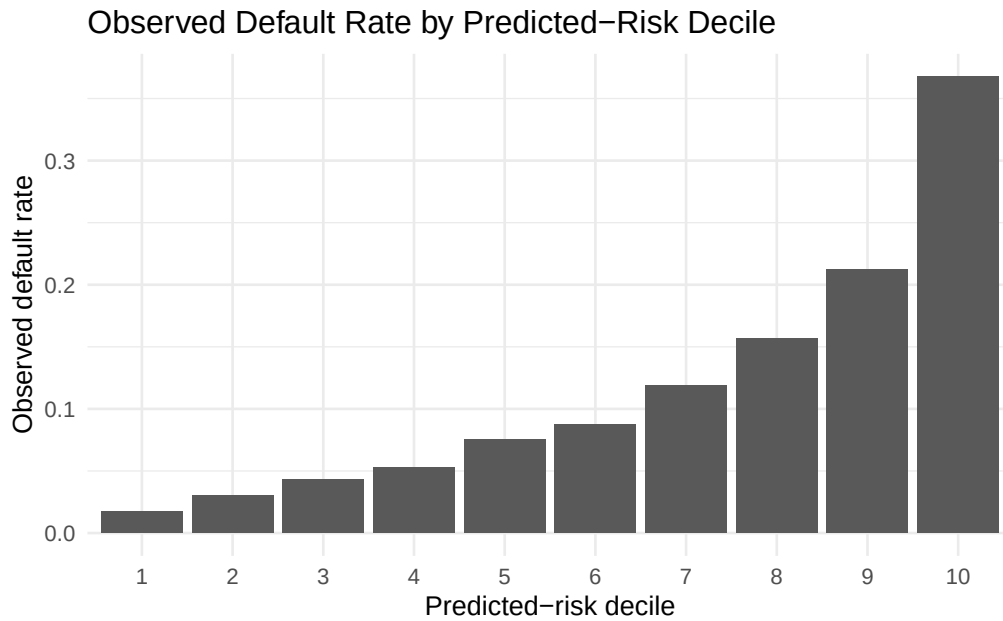
Quantity	Estimate	Reference
Calibration-in-the-large intercept	0.0093	0
Calibration model intercept	0.0073	0
Calibration slope	0.9989	1

These quantities should be interpreted together with the calibration curve and Brier/log-loss results rather than as stand-alone pass/fail tests.

Table 14: Observed outcomes across predicted-risk deciles.

Decile	N	MeanPredicted	ObservedDefaultRate	Defaults	ShareOfDefaults
1	3831	0.0170	0.0175	67	0.0151
2	3831	0.0304	0.0303	116	0.0261
3	3831	0.0425	0.0431	165	0.0371
4	3831	0.0558	0.0525	201	0.0452
5	3831	0.0718	0.0752	288	0.0647
6	3830	0.0911	0.0875	335	0.0753
7	3830	0.1161	0.1191	456	0.1025
8	3830	0.1511	0.1567	600	0.1348
9	3830	0.2077	0.2125	814	0.1829
10	3830	0.3699	0.3676	1408	0.3164

6.6 Risk Stratification



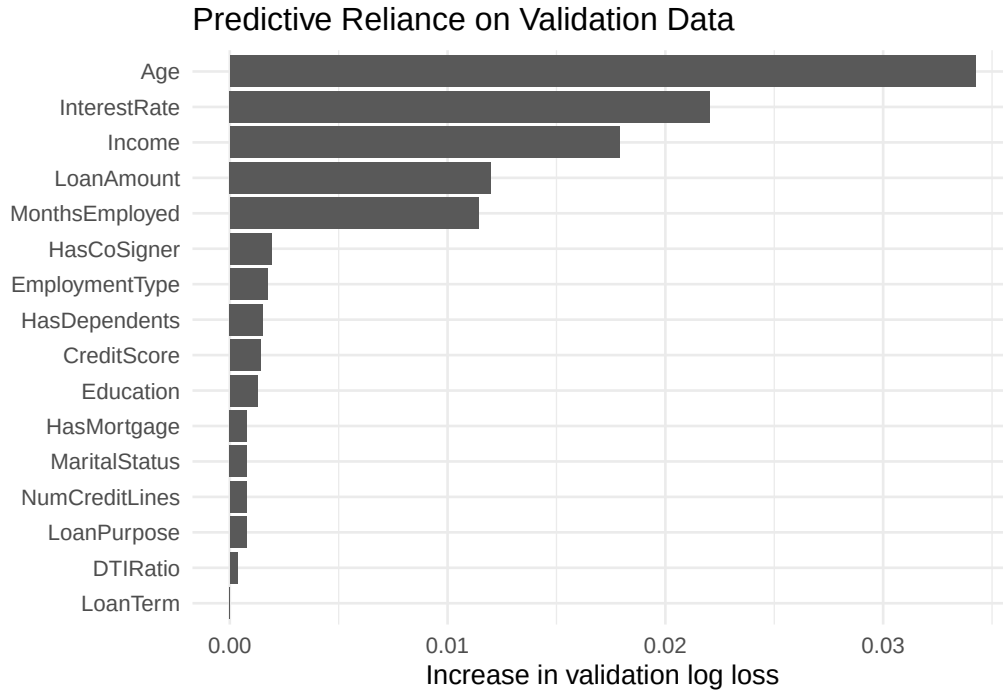
6.7 Predictor Reliance

Permutation importance evaluates how much validation log loss increases when one predictor is randomly disrupted.

This measures **predictive reliance**, not causal effect.

Table 15: Validation permutation importance.

Feature	IncreaseInLogLoss
Age	0.03423
InterestRate	0.02205
Income	0.01790
LoanAmount	0.01199
MonthsEmployed	0.01141
HasCoSigner	0.00191
EmploymentType	0.00173
HasDependents	0.00152
CreditScore	0.00142
Education	0.00127
HasMortgage	0.00080
MaritalStatus	0.00079
NumCreditLines	0.00076
LoanPurpose	0.00076
DTIRatio	0.00036
LoanTerm	0.00000



6.8 Age-Related Subgroup and Sensitivity Diagnostics

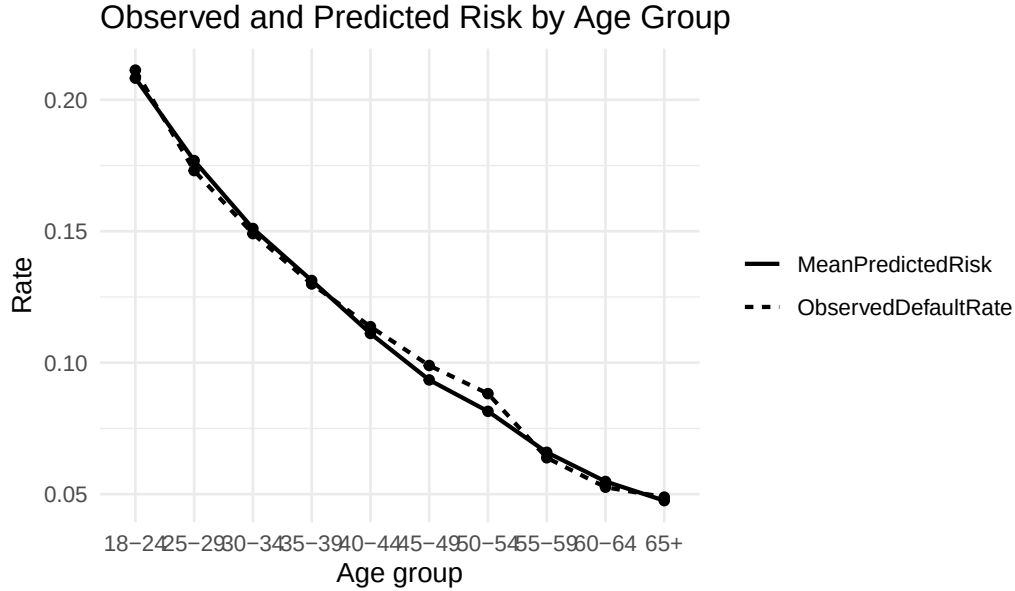
Previous analysis found that Age can be highly influential. Therefore, Age is examined separately as a diagnostic rather than interpreted causally.

Table 16: Age-group observed risk and calibration diagnostic on the held-out test set.

AgeGroup	Loans	Defaults	ObservedDefaultRate	MeanPredictedRisk	CalibrationGap
18-24	5017	1060	0.2113	0.2082	0.0030
25-29	3600	623	0.1731	0.1769	-0.0038
30-34	3736	557	0.1491	0.1510	-0.0020
35-39	3794	493	0.1299	0.1313	-0.0013
40-44	3597	409	0.1137	0.1111	0.0026
45-49	3740	370	0.0989	0.0934	0.0055
50-54	3627	320	0.0882	0.0815	0.0067
55-59	3807	243	0.0638	0.0659	-0.0021
60-64	3724	196	0.0526	0.0548	-0.0022
65+	3663	179	0.0489	0.0475	0.0013

Table 17: Age-group classification behavior at the selected review threshold.

AgeGroup	FlaggedRate	Recall	Specificity
18-24	0.7257	0.8934	0.3192
25-29	0.6242	0.8395	0.4209
30-34	0.5145	0.7846	0.5329
35-39	0.4257	0.6978	0.6150
40-44	0.3228	0.6088	0.7139
45-49	0.2420	0.5243	0.7890
50-54	0.1817	0.4500	0.8443
55-59	0.1271	0.3827	0.8903
60-64	0.0806	0.3265	0.9331
65+	0.0631	0.2793	0.9480



Large differences across age groups should be treated as a model diagnostic requiring interpretation and, where appropriate, sensitivity analysis. They are not evidence that Age has a causal effect on default.

The age-group table is a **limited subgroup diagnostic**, not a formal fairness determination. Age is strongly associated with the model score in these data, so differences in observed risk, calibration, recall, specificity, and flagging are reported transparently. A complete fairness audit would require a clear legal and decision context, additional protected-group variables, appropriate fairness definitions, and analysis of the consequences of the decision rule.

6.9 Separate Validation / Stress Test

Table 18: Probability performance on the separate stress-test data.

N	DefaultRate	LogLoss	Brier	ROC_AUC	PR_AUC
5000	0.1534	0.3652	0.1103	0.7583	0.4312

Table 19: Operating characteristics on the separate stress-test data.

Threshold	Recall	Precision	Specificity	FalsePositiveRate	FlaggedRate	Accuracy
0.1175	0.7249	0.2645	0.6348	0.3652	0.4204	0.6486

The additional validation file is used as a stress test. Its synthetic construction means that it does not constitute independent external validation. It is included to examine how the fixed model behaves under a deliberately shifted population.

6.10 Predictive Stability Under Controlled Distribution Shift

The empirical results above establish the internally validated baseline. The following results examine what happens to the same statistical properties when the deployment distribution is deliberately changed.

6.10.1 Covariate shift under correct specification

Under the correctly specified logistic model, raw log loss increases as δ increases, but the oracle loss increases almost identically because the event distribution and composition of the deployment population also change.

Table 20: Pure covariate shift under correct model specification.

delta	event_rate	model_log_loss	oracle_log_loss	excess_log_loss	model_brier	oracle_brier	probability_mse
0.0	0.1201	0.3169	0.3165	0.0004	0.0934	0.0933	0.0001
0.5	0.1623	0.3812	0.3806	0.0006	0.1175	0.1173	0.0002
1.0	0.2142	0.4444	0.4436	0.0008	0.1424	0.1421	0.0003
1.5	0.2758	0.5026	0.5015	0.0011	0.1659	0.1655	0.0004
2.0	0.3462	0.5498	0.5483	0.0015	0.1855	0.1849	0.0006

Probability MSE remains small, increasing from approximately 0.00012 to 0.00062. The fitted model therefore remains close to the oracle even as the population changes substantially. This control condition shows that raw predictive loss should not be interpreted without considering the changing deployment problem itself.

6.10.2 Covariate shift exposes model misspecification

When the true outcome mechanism contains an omitted quadratic term, the effect of covariate shift changes substantially.

Table 21: Probability deterioration under covariate shift and model misspecification.

beta4	delta	event_rate	probability_mse	excess_log_loss	excess_brier
0.00	0.0	0.1199	0.0001	0.0004	0.0001
0.00	0.5	0.1622	0.0002	0.0005	0.0002
0.00	1.0	0.2139	0.0003	0.0008	0.0003
0.00	1.5	0.2758	0.0004	0.0011	0.0004
0.00	2.0	0.3456	0.0006	0.0015	0.0006
0.25	0.0	0.1204	0.0012	0.0053	0.0012
0.25	0.5	0.1779	0.0026	0.0072	0.0026
0.25	1.0	0.2634	0.0059	0.0147	0.0058
0.25	1.5	0.3757	0.0119	0.0301	0.0119
0.25	2.0	0.5063	0.0206	0.0540	0.0206
0.50	0.0	0.1201	0.0043	0.0221	0.0043
0.50	0.5	0.1882	0.0080	0.0240	0.0079
0.50	1.0	0.2955	0.0164	0.0436	0.0164
0.50	1.5	0.4349	0.0293	0.0778	0.0292
0.50	2.0	0.5908	0.0442	0.1224	0.0441

At $\delta = 2$, probability MSE is approximately 0.00062 under correct specification but approximately 0.04422 under strong misspecification. The shift therefore magnifies an approximation error that is much smaller near the development population.

The calibration and discrimination results reveal a second feature.

Table 22: Discrimination and calibration under covariate shift.

beta4	delta	model_auc	oracle_auc	cal_intercept	cal_slope	probability_mse
0.00	0.0	0.7624	0.7629	0.0043	1.0014	0.0001
0.00	0.5	0.7597	0.7602	0.0045	0.9992	0.0002
0.00	1.0	0.7579	0.7584	0.0091	1.0021	0.0003
0.00	1.5	0.7556	0.7561	0.0104	1.0022	0.0004
0.00	2.0	0.7537	0.7541	0.0094	0.9999	0.0006
0.25	0.0	0.7829	0.7892	0.0077	1.0044	0.0011
0.25	0.5	0.8077	0.8116	0.2285	1.1495	0.0025
0.25	1.0	0.8228	0.8278	0.4101	1.2682	0.0058
0.25	1.5	0.8339	0.8413	0.5682	1.3695	0.0118
0.25	2.0	0.8437	0.8540	0.7188	1.4559	0.0204
0.50	0.0	0.7873	0.8174	-0.0077	0.9918	0.0043
0.50	0.5	0.8420	0.8530	0.4661	1.3171	0.0079
0.50	1.0	0.8651	0.8753	0.7711	1.5401	0.0162
0.50	1.5	0.8771	0.8904	1.0009	1.6905	0.0292
0.50	2.0	0.8883	0.9044	1.2132	1.8097	0.0440

Under strong misspecification, model ROC-AUC increases from approximately 0.787 to 0.888 as δ moves from 0 to 2, while probability MSE increases from approximately 0.0043 to 0.0440 and the calibration slope increases from approximately 0.99 to 1.81. The deployment population becomes easier to rank while the numerical probabilities become less accurate.

6.10.3 Decision-threshold stability

The optimal balanced-accuracy threshold moves systematically with the deployment population.

Optimal Decision Thresholds Under Covariate Shift

Thresholds maximizing balanced accuracy across deployment populations

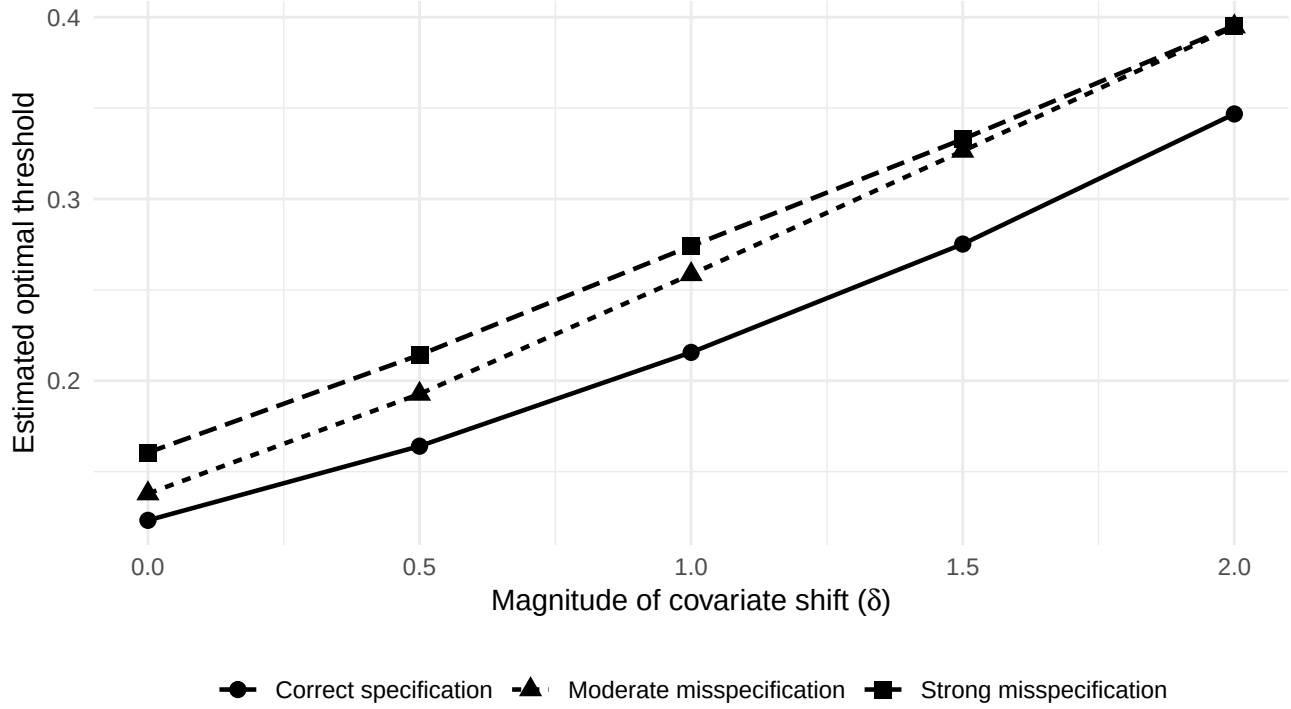


Table 23: Decision-threshold movement and balanced-accuracy performance under covariate shift.

beta4	delta	Baseline threshold	Shifted threshold	Threshold change	Frozen BA	Adapted BA	BA gain
0.00	0.0	0.1222	0.1232	0.0009	0.6912	0.6911	-0.0002
0.00	0.5	0.1222	0.1640	0.0418	0.6808	0.6911	0.0103
0.00	1.0	0.1222	0.2156	0.0934	0.6518	0.6879	0.0360
0.00	1.5	0.1222	0.2752	0.1529	0.6159	0.6866	0.0707
0.00	2.0	0.1222	0.3468	0.2246	0.5794	0.6850	0.1055
0.25	0.0	0.1374	0.1379	0.0004	0.7112	0.7111	-0.0001
0.25	0.5	0.1374	0.1927	0.0552	0.7187	0.7325	0.0137
0.25	1.0	0.1374	0.2585	0.1211	0.6949	0.7446	0.0497
0.25	1.5	0.1374	0.3264	0.1890	0.6549	0.7527	0.0978
0.25	2.0	0.1374	0.3948	0.2574	0.6130	0.7610	0.1480
0.50	0.0	0.1600	0.1605	0.0005	0.7285	0.7284	-0.0001
0.50	0.5	0.1600	0.2142	0.0542	0.7558	0.7675	0.0118
0.50	1.0	0.1600	0.2741	0.1142	0.7379	0.7843	0.0464
0.50	1.5	0.1600	0.3329	0.1729	0.6993	0.7930	0.0938
0.50	2.0	0.1600	0.3954	0.2354	0.6560	0.8033	0.1472

Even under correct specification, the mean optimal threshold rises from approximately 0.122 at baseline to 0.347 at $\delta = 2$. Under strong misspecification, the threshold rises from approximately 0.160 to 0.395. At $\delta = 2$, the frozen threshold produces mean balanced accuracy of approximately 0.656 compared with approximately 0.803 after threshold adaptation, a gain of approximately 0.147 with MCSE approximately 0.00114.

6.10.4 Intercept outcome shift: calibration can fail while ranking remains stable

Calibration Under Intercept Outcome Shift

Simulation estimates closely follow the theoretical relationship

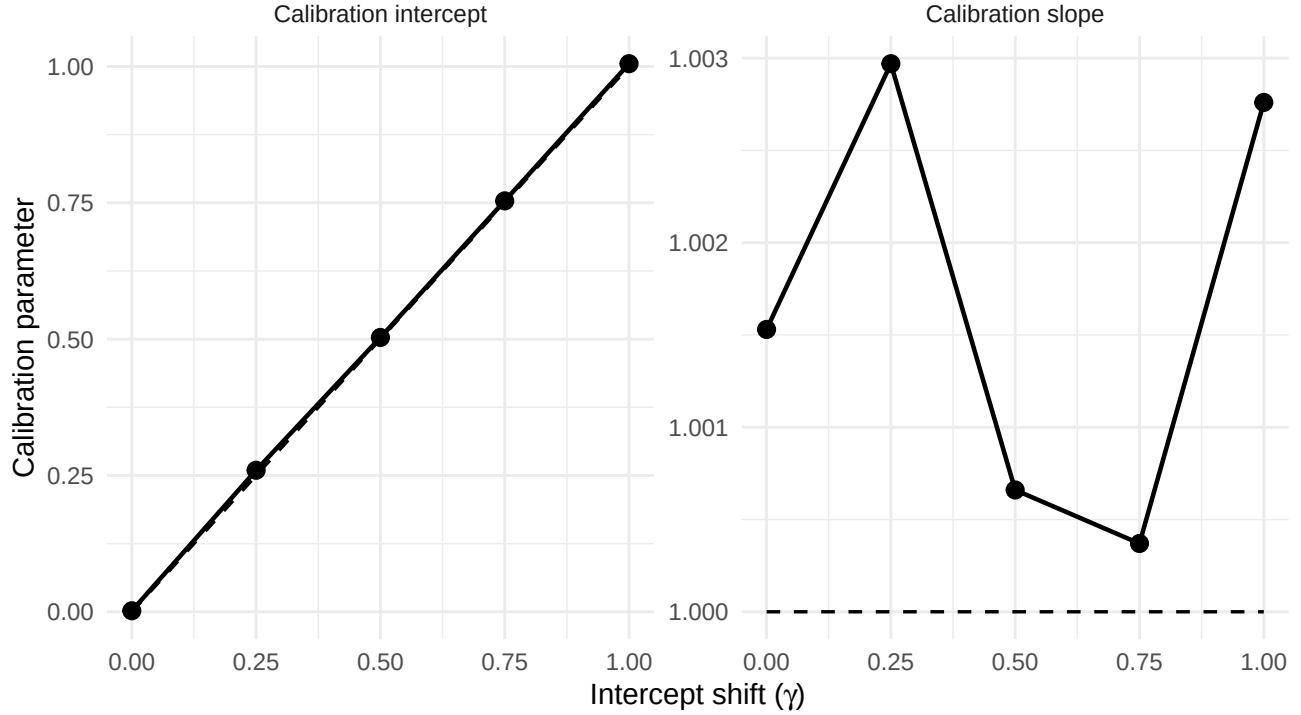


Table 24: Intercept outcome shift: event rate and discrimination.

Intercept shift (γ)	Event rate	Model AUC	Oracle AUC	AUC gap
0.00	0.1198	0.7622	0.7627	0.0005
0.25	0.1453	0.7606	0.7611	0.0005
0.50	0.1739	0.7588	0.7593	0.0005
0.75	0.2069	0.7570	0.7575	0.0005
1.00	0.2429	0.7564	0.7569	0.0005

Table 25: Intercept outcome shift: calibration and probability accuracy.

Intercept shift (γ)	Calibration intercept	MCSE: intercept	Calibration slope	MCSE: slope	Probability MSE	Excess log loss
0.00	0.0019	0.0058	1.0015	0.0031	0.0001	0.0004
0.25	0.2596	0.0057	1.0030	0.0030	0.0010	0.0038
0.50	0.5031	0.0059	1.0007	0.0030	0.0042	0.0148
0.75	0.7536	0.0057	1.0004	0.0029	0.0104	0.0356
1.00	1.0052	0.0056	1.0028	0.0029	0.0201	0.0675

The calibration intercept closely tracks the imposed value of γ , while the calibration slope remains approximately one. At $\gamma = 1$, the mean calibration intercept is approximately 1.005 and the mean slope is approximately 1.003. Probability MSE rises from approximately 0.00011 to 0.02010, yet the model–oracle AUC gap remains about 0.0005. The model therefore retains ranking information while its absolute probabilities become systematically wrong.

6.10.5 Slope outcome shift: changing relationships damage discrimination

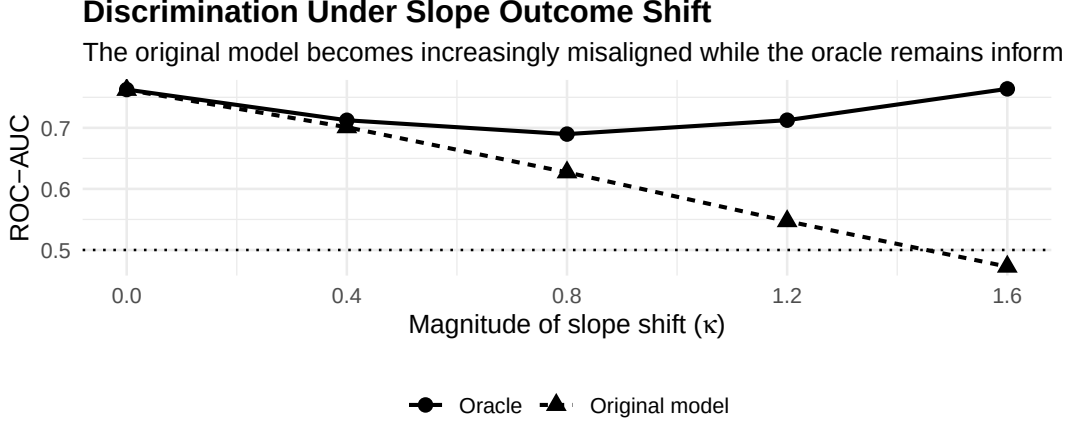


Table 26: Slope outcome shift: discrimination.

Slope shift (kappa)	Deployment beta1	Event rate	Model AUC	MCSE: model AUC	Oracle AUC	AUC gap	MCSE: AUC gap
0.0	0.8	0.1199	0.7622	0.0004	0.7627	0.0005	0.0000
0.4	0.4	0.1201	0.7010	0.0005	0.7124	0.0114	0.0002
0.8	0.0	0.1199	0.6271	0.0006	0.6898	0.0626	0.0006
1.2	-0.4	0.1201	0.5472	0.0007	0.7126	0.1654	0.0008
1.6	-0.8	0.1196	0.4727	0.0008	0.7638	0.2911	0.0009

Table 27: Slope outcome shift: calibration and probability accuracy.

Slope shift (kappa)	Calibration intercept	Calibration slope	Probability MSE	Excess log loss	MCSE: excess log loss
0.0	0.0019	1.0016	0.0001	0.0004	0.0000
0.4	-0.4903	0.7188	0.0019	0.0075	0.0001
0.8	-1.0417	0.4326	0.0075	0.0312	0.0002
1.2	-1.6283	0.1574	0.0161	0.0705	0.0004
1.6	-2.2186	-0.0905	0.0267	0.1191	0.0005

With event prevalence held near 12%, the original model’s ROC-AUC falls from approximately 0.762 to 0.473 as the true effect of X_1 weakens and reverses. At the strongest shift, the oracle ROC-AUC is approximately 0.764, producing an AUC gap of approximately 0.291 with MCSE approximately 0.00093. The calibration slope falls from approximately 1.00 to -0.09 , probability MSE increases to approximately 0.02672, and excess log loss reaches approximately 0.11914.

The oracle remains informative because the prediction problem itself still contains signal; the original model fails because the relationship it learned is no longer the deployment relationship.

6.10.6 Integrated result

The experiments do not produce one universal pattern of model deterioration. Covariate shift can expose misspecification and alter operating thresholds; intercept outcome shift can damage calibration while preserving ranking; and slope outcome shift can damage both calibration and discrimination. The common implication is that stable or improving ROC-AUC does not guarantee stable probabilities, calibration, or decision rules.

7 Discussion

This study began with a practical loan-default prediction problem but leads to a broader statistical conclusion about predictive inference under distributional change. The empirical analysis establishes a model that performs reasonably well under internal validation. The simulation analysis then shows why that evidence should not be interpreted as proof that the same probabilities, calibration, ranking, or decision threshold will remain appropriate after deployment.

The empirical model-selection exercise is therefore a supporting component rather than the final research question. Raw logistic regression achieved the lowest validation log loss among the candidate models, with performance very close to Platt-calibrated logistic regression. The more flexible random-forest and boosting candidates did not improve probability estimation under the prespecified validation criterion in this dataset. On the locked test set, the selected model achieved ROC-AUC 0.7585, PR-AUC 0.3288, Brier score 0.0908, and log loss 0.3111. Its calibration intercept and slope were close to the ideal values, providing evidence that the probability scale was appropriate for the held-out observations drawn from the same source.

The threshold analysis adds a separate decision layer. Balanced accuracy selected a validation threshold of 0.1175, which produced test recall of approximately 0.684 and specificity of approximately 0.701. This threshold is not an intrinsic property of logistic regression and is not an economically optimal lending cutoff. It is the solution to a particular statistical operating objective under the validation population.

The covariate-shift experiments explain why internal success may not transfer unchanged to deployment. When the conditional logistic model was correctly specified, moving the distribution of X_1 produced only small excess probability error relative to the oracle. This is an important control result: a change in $P(X)$ does not automatically invalidate a correct model of $P(Y | X)$. Raw loss can increase simply because the composition and event distribution of the population have changed.

The result changes when the fitted model is misspecified. Omitting a quadratic effect creates an approximation error that is relatively modest near the development distribution but becomes much larger as deployment moves toward regions where the omitted curvature matters more. This demonstrates an interaction between covariate shift and model misspecification rather than a universal failure of logistic regression under shift.

The same experiment also demonstrates why discrimination should not be treated as a complete measure of predictive validity. Under strong misspecification, ROC-AUC increased as covariate shift became stronger even though probability MSE and calibration error deteriorated substantially. The deployment population became easier to rank, but the numerical probabilities became less trustworthy. AUC therefore answered the ranking question correctly while failing to reveal a different probability-estimation problem.

The intercept-shift experiment isolates that distinction further. The estimated calibration intercept closely followed the imposed shift, the calibration slope remained near one, and the model-oracle AUC gap remained negligible. The old model continued to rank observations effectively but systematically underestimated the new level of risk. In a setting approximating this structure, recalibration may be sufficient because the ordering information has largely survived.

The slope-shift experiment represents a fundamentally different failure. When the true effect of X_1 weakened and eventually reversed, the original model’s ranking deteriorated even though the oracle remained informative. At the strongest shift, the original ROC-AUC fell below 0.50 while the oracle remained near 0.764. This is not a problem that an intercept correction alone can solve because the predictor–outcome relationship itself has changed.

Decision-threshold stability provides a final distinction. Even under correct model specification, the balanced-accuracy-optimal threshold moved substantially after covariate shift. Thus model validity and decision-rule validity are not identical. A model may still estimate useful conditional probabilities while the operating threshold chosen for the development population becomes suboptimal for a new deployment population.

These results change how the empirical loan model should be interpreted. The test-set results establish internal predictive performance under the observed data source. They do not establish that a future borrower population will have the same predictor distribution, baseline default risk, conditional predictor effects, calibration, or optimal review threshold. The simulation does not prove that these shifts occur in the empirical data; rather, it identifies the statistical consequences that would follow if different forms of shift occurred.

For deployment monitoring, the findings argue against relying on ROC-AUC alone. A probability-based risk system should separately monitor input distributions, event prevalence, calibration intercept and slope, probability losses such as Brier score or log loss once outcomes mature, ranking measures, score distributions, review rates, and threshold operating characteristics. The appropriate intervention depends on the observed failure mode. Intercept recalibration, broader recalibration, threshold revision, and complete model redevelopment are statistically different responses to different problems.

The central result can therefore be stated compactly:

Stable discrimination does not guarantee stable probabilities, calibration, or decisions under distribution shift.

That conclusion is broader than the specific loan dataset. It concerns the deployment of probability models whenever the future population may differ from the population used for development.

8 Limitations

Several limitations constrain interpretation of the empirical and simulation evidence.

First, the empirical `Default` response has no documented observation horizon. The fitted probabilities therefore concern the recorded default outcome rather than a verified 12-month, lifetime, or regulatory probability of default.

Second, the empirical data contain no loan-origination or outcome dates. The train/calibration/validation/test design is stratified rather than chronological, so the study cannot establish performance on a genuinely later origination cohort.

Third, the data do not contain a borrower identifier that establishes whether the same borrower can appear more than once. The analysis therefore assumes independent loan records. If repeated borrowers exist, borrower-grouped splitting would be necessary.

Fourth, the dataset contains no longitudinal repayment history, days-past-due trajectory, recoveries, exposure at default, or loss given default. The empirical model is therefore a binary recorded-default risk model rather than a time-to-default, early-warning, or expected-loss model.

Fifth, the separate validation file is synthetic and functions only as a stress test. It does not constitute genuine external validation and cannot establish transportability to an independent lending population.

Sixth, balanced accuracy is a statistical threshold objective. The available data do not contain verified costs of false negatives and false positives, review-capacity constraints, revenues, recoveries, EAD, or LGD required to establish an economically optimal threshold.

Seventh, the subgroup analysis is descriptive and incomplete. The available variables do not support a comprehensive fairness audit, and the study does not make a fairness determination.

The simulation study is deliberately stylized. It uses three independent normally distributed predictors and logistic data-generating mechanisms so that specific forms of shift can be isolated. Real lending populations may contain correlated predictors, interactions, missingness, policy effects, temporal dependence, and several simultaneous forms of distribution change.

The simulated values of δ , β_4 , γ , and κ are experimental design conditions, not empirical estimates of real credit-market drift. The simulation establishes mechanisms under controlled assumptions; it does not establish that any particular shift occurred in the observed loan dataset.

Finally, the Monte Carlo standard errors quantify simulation precision, not uncertainty about real-world future performance. Genuine claims about deployment transportability require temporal or external data.

9 Future Research

The most important next step is empirical validation under real distributional change. A dataset with documented origination dates, outcome dates, a defined default horizon, and later-period cohorts would permit direct out-of-time evaluation of calibration, discrimination, and threshold stability. Data from an independent lender or lending environment would provide an even stronger transportability test.

A second extension is adaptation after detected shift. The present study characterizes several failure modes; future work should compare intercept-only recalibration, recalibration of both intercept and slope, threshold revision, importance weighting under covariate shift, and complete model refitting. Each adaptation method should be estimated on a dedicated adaptation sample and evaluated on a separate shifted holdout sample.

A richer credit-risk application should incorporate longitudinal repayment histories, exposure at default, loss given default, recoveries, review costs, and revenues. Those quantities would allow the operating rule to be chosen using expected loss or utility rather than balanced accuracy.

Finally, the simulation framework can be extended across sample size, baseline prevalence, signal strength, correlated predictors, alternative nonlinearities, interactions, and simultaneous changes in $P(X)$ and $P(Y | X)$. Such work would move from illustrating distinct failure mechanisms toward identifying conditions under which particular adaptation strategies are statistically reliable.

10 Reproducibility and Software

The empirical workflow is implemented in R Markdown with a fixed development design, training-only preprocessing, saved model objects, and a validation-selected threshold. The associated Shiny application uses the same saved probability model, preprocessing recipe, calibration choice, and threshold, and is intended as a research and decision-support interface rather than an automated lending-decision system.

The distribution-shift experiments use 500 Monte Carlo replications per condition with simulation seed 1998. The manuscript records the frozen simulation summaries and reconstructs the reported tables and figures from those results. Monte Carlo standard errors are reported where they are central to the interpretation.

Code is hidden in the rendered manuscript so that the PDF emphasizes the statistical design, results, and interpretation while the R Markdown source preserves the computational workflow.

11 Conclusion

This study evaluates loan-default prediction as a probability-based system that must remain statistically meaningful beyond the development sample. The empirical analysis establishes the internal baseline: raw logistic regression achieved the lowest validation log loss among the candidate models, produced useful held-out discrimination, remained close to ideal calibration, and supported a validation-selected balanced-accuracy threshold of 0.1175.

The central contribution comes from asking what happens to those same properties when the deployment distribution changes. Pure covariate shift produced little excess probability error when the conditional model was correctly specified, but the combination of covariate shift and model misspecification produced substantial calibration and probability deterioration. Intercept outcome shift altered absolute risk while largely preserving ranking. Slope outcome shift changed the predictor–outcome relationship itself and eventually destroyed the original model’s discrimination. The operating threshold also moved substantially across deployment populations, including under correct specification.

The broader conclusion is that **distribution shift does not create one universal form of predictive failure**. Different changes in the data-generating process affect different parts of the predictive system. A model can retain or improve discrimination while its probabilities become miscalibrated, remain probabilistically useful while its decision threshold becomes inappropriate, or lose ranking ability when predictor effects change.

Accordingly, internal test performance should be treated as evidence about the development environment rather than as proof of future stability. Probability accuracy, calibration, discrimination, and decision performance should be evaluated and monitored as distinct but connected properties of a deployed predictive model.

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